ALK-positive anaplastic large cell lymphoma initially diagnosed as neurosarcoidosis in a 12-year-old girl

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We present a rare case of optic neuropathy due to anaplastic lymphoma kinase–positive (ALK+) anaplastic large cell lymphoma (ALCL) with optic nerve infiltration in a 12-year-old girl who presented with acute unilateral vision loss, diplopia, and headache after two prior hospitalizations at an outside facility for disk edema. She had a presumptive diagnosis of neurosarcoidosis and empiric treatment had been initiated with high-dose corticosteroids. Ongoing worsening of vision prompted presentation at our facility. Histopathological examination from a cervical lymph node biopsy revealed ALK+ ALCL with central nervous system involvement. The patient's vision returned after initiation of chemotherapy. This case highlights the importance of considering malignancy in the differential for optic neuritis.

Anaplastic large cell lymphoma (ALCL) is a rare type of T cell non-Hodgkin lymphoma (NHL) that accounts for approximately 3%-5% of all NHLs.[1] Neurolymphomatosis is defined as the invasion of peripheral nerves by malignant lymphomatous cells and has been found in only 2.5%-10% of T cell lymphoma cases.[2,3] This report details a rare case of optic neurolymphomatosis due to anaplastic lymphoma kinase–positive (ALK+) ALCL that was diagnosed after presentation with acute optic neuropathy.

Case Report

A 12-year-old previously healthy Black girl presented to an outside facility with acute vision loss in her left eye, diplopia, and headache. She was found to have an abducens nerve palsy, bilateral disk edema, and high intracranial pressure. Magnetic resonance imaging (MRI) at that time showed diffuse leptomeningeal enhancement of the cerebral and cerebellar hemispheres with abnormal signal changes in the subarachnoid space and a right hyperintense parietal lesion. Cerebrospinal fluid (CSF) studies showed an elevated protein level of 298, increased white blood cell count of 414 (with 66% lymphocytes), and a low glucose level of 28. CSF meningitis and encephalitis panels and cultures were negative. She was presumptively diagnosed with viral meningitis and discharged on acetazolamide. She was admitted a second time for continued symptoms, and repeat MRI showed worsening leptomeningeal enhancement with new extension along the optic nerve sheaths bilaterally. Repeat lumbar puncture revealed an opening pressure of 49 mm Hg, and cytology studies showed atypical cells. She was diagnosed with presumed neurosarcoidosis based on elevated CSF angiotensin converting enzyme level and hilar adenopathy on computed tomography of the chest. She was discharged home after intravenous corticosteroid therapy on tapering oral corticosteroids and acetazolamide.

Several weeks later, she presented at Children's Hospital New Orleans with best-corrected visual acuity of 20/25 in the right eye and 20/200 in the left eye, with a new 3+ relative afferent pupillary defect on the left. Her prior disk edema had resolved, and there was no ophthalmoplegia. There was a left inferior quadrantanopia and dyschromatopsia. Given her recent history of presumed neurosarcoidosis, she was admitted for high-dose corticosteroids. Lymph node biopsy was not performed prior to initiation of corticosteroids. Visual acuity in the left eye declined to counting fingers 2 days after initiation of high-dose corticosteroids. Repeat imaging showed worsening enhancement involving the intracranial portions of the bilateral optic nerves, optic chiasm, and optic tracts as well as a hyperintense lesion in the right parietal lobe (Figures 1

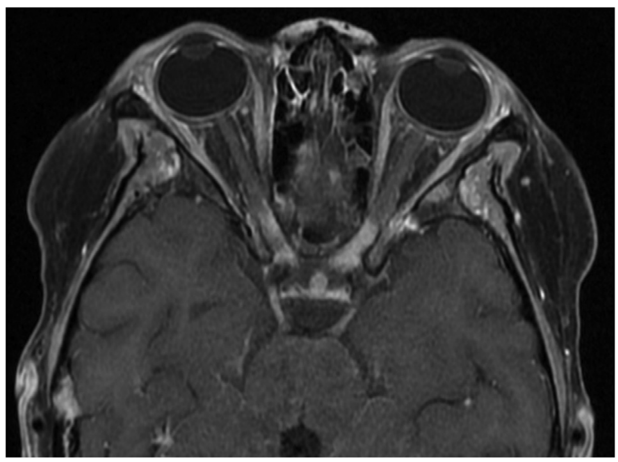

FIG 1. Magnetic resonance imaging (MRI) of the brain and orbits (axial cut, T1-weighted, with fat suppression post contrast) on presentation, notable for thickening and enhancement of the intracranial portions of the bilateral optic nerves with chiasmal involvement.

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and 2). Review of systems were negative other than recent documented weight loss of 45 pounds.

The lack of clinical response to high-dose corticosteroids, significant weight loss, and absence of chitotriosidase elevation brought into question the diagnosis of neurosarcoidosis and prompted consultation of rheumatology, oncology, and neurology services. The patient underwent biopsy of an enlarged cervical lymph node, which was positive for ALK+ ALCL (Figures 3 and 4). Subsequent bone marrow

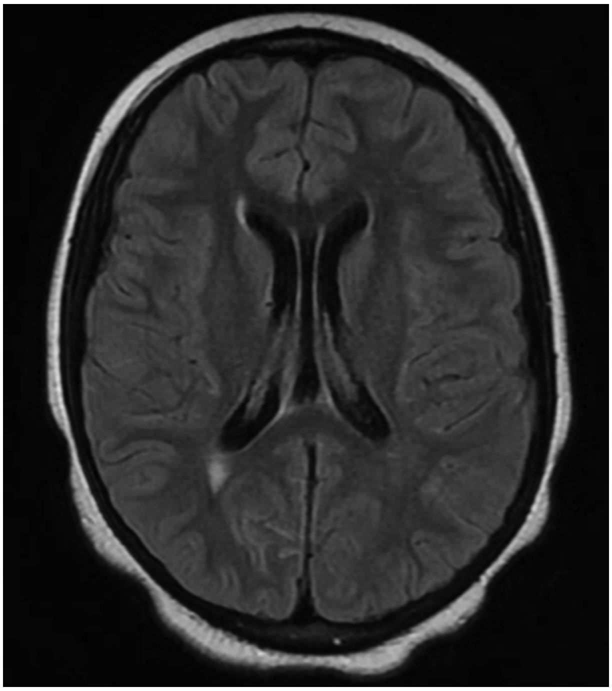

FIG 2. MRI of the brain and orbits (axial cut, T2-FLAIR) on presentation to our institution, notable for a hyperintense lesion in the right parietal lobe.

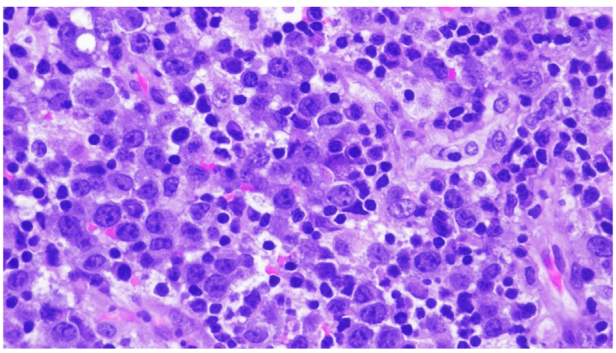

FIG 3. Left cervical lymph node from our patient following excisional biopsy showing diffuse distribution of numerous large atypical cells with abundant cytoplasm and round-to-oval nuclei with coarse chromatin and variable nucleoli; some forms are multinucleate cells with abundant clear to pale cytoplasm (hematoxylin and eosin, original magnification ×400). Numerous mitotic figures and apoptoses are seen. The reactive background shows a mixed population of lymphocytes with no increased eosinophils.

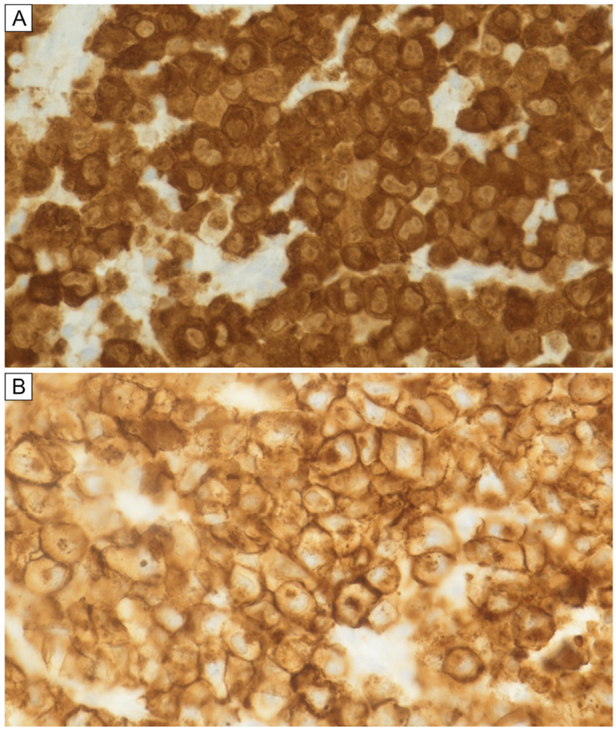

FIG 4. Left cervical lymph node from our patient status after excisional biopsy. Neoplastic cells display positive ALK expression (A) and positive CD30 expression (B) diffusely (original magnification ×400).

biopsy and lumbar puncture revealed malignant infiltration of bone marrow and CSF consistent with stage IV disease. Systemic and intrathecal chemotherapy was initiated per Children's Oncology Group protocol ANHL12P1.[4] Four days after starting chemotherapy, her visual acuity returned to 20/30 in her left eye, without relative afferent pupillary defect, and the dyschromatopsia resolved; however, the left inferior quadrantanopia persisted. The patient received a total of 6 cycles of chemotherapy per ANHL12P1, with additional intrathecal chemotherapy at the start of each cycle.

Repeat MRI performed after cycle 2 showed a complete response, with resolution of the previously seen intracranial enhancements and only mild residual thickening and enhancement of the optic nerves (Figure 5). Repeat bone marrow and CSF analysis were negative for malignancy, and cytology studies were negative for malignant cells. Opening pressures were obtained with each lumbar puncture, and though improved, they remained elevated through much of her treatment course. Her treatment was consolidated with craniospinal radiation. She is now off chemotherapy, with no evidence of recurrent disease. Six weeks following completion of chemotherapy, visual acuity was 20/20 in both eyes, without relative afferent pupillary defect and no dyschromatopsia. The patient's congruous left inferior quadrantanopia was still present on automated visual field and correlated with a right parietal lesion found on prior MRI scans. The patient was also

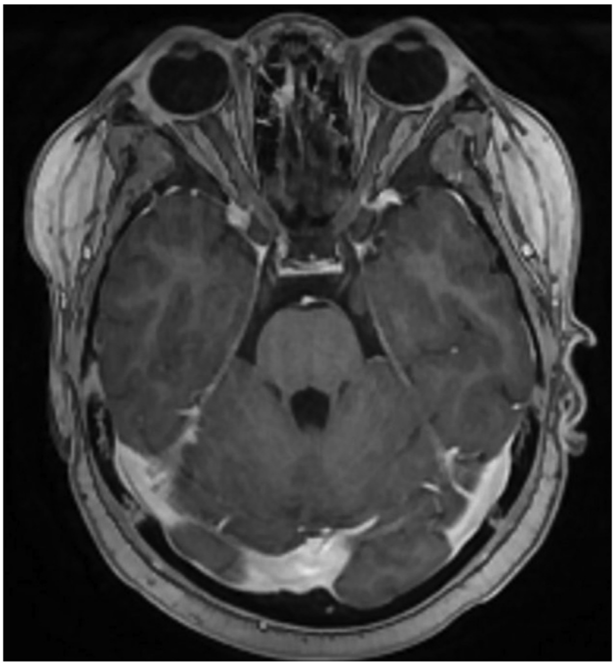

FIG 5. MRI of the brain and orbits (axial cut, T1-weighted post-contrast) on completion of therapy showing resolution of optic nerve enhancement.

found to have bilateral temporal retinal nerve fiber layer on peripapillary optical coherence tomography (OCT) and ganglion cell layer thinning on macular OCT, more prominent of the left, suggesting possible early injury to the anterior visual pathway. Funduscopy was generally unremarkable.

Discussion

Cases of reversible vision loss with treatment of ALK+ ALCL are sparse in the literature, and to our knowledge, this is a unique case of ALK+ ALCL diagnosed after initial presentation of acute optic neuropathy. The patient presented with a prior diagnosis of presumed neurosarcoidosis, but her vision declined with high-dose corticosteroid therapy. Subsequent biopsy of an enlarged cervical lymph node revealed ALK+ ALCL. She exhibited radiographic evidence of optic neurolymphomatosis as well as positive CSF and bone marrow cytology.

In contrast to ALK− ALCL, ALK+ ALCL has been associated with more favorable outcomes, with overall survival rates of 70%-90%.[5] ALCL is classically a chemosensitive disease, and treatment with the combination of chemotherapy and immunotherapy has contributed to the high survival rates.[6] Only a few cases of central nervous system (CNS) involvement in ALK+ ALCL have been published.[7,8] Literature states that prognostic factors in CNS disease are similar to those of systemic ALCL, with lower mortality seen in younger patients and in those with ALK positivity, although it has been suggested that CNS disease demonstrates more aggressive behavior.[2]

The International Primary CNS Lymphoma Collaborative Group retrospectively analyzed 50 patients assembled from 12 centers in 5 countries over a 16-year period and found that neurolymphomatosis was the initial manifestation of malignancy in only 26% of cases, and of these cases, only 46% had cranial nerve involvement. Response to treatment was only observed in 46% of total cases.[2] Permanent blindness in the setting of ALK-positive ALCL with optic neurolymphomatosis has been reported.[8] Our patient experienced visual recovery after initiation of chemotherapy.

It is important to acknowledge several missed opportunities for correct diagnosis in this case, including lack of tissue biopsy prior to diagnosing neurosarcoid, significant weight loss, and abnormal CSF studies and cytology. A thorough review of outside hospital records may have led to a more prompt diagnosis. This case underscores the importance of a comprehensive diagnostic approach in pediatric patients with acute vision loss and demonstrates the significance of obtaining pathological confirmation before initiating empiric corticosteroid treatment when malignancy is a consideration. Unusual findings should prompt further investigation and collaboration between specialties is crucial in achieving the best treatment outcomes.

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